Experiment No : 2

Title:

Use var, let, const, and demonstrate basic features like template literals and destructuring; build an Online Shopping Website with user input.

Software/Tools Required:

- Visual Studio Code (or any text editor)
- Web Browser (Google Chrome, Microsoft Edge, Mozilla Firefox, etc.)
- Basic knowledge of HTML5 and modern JavaScript (ES6+) and CSS.

Theory:

JavaScript provides different ways to declare variables, each with its own scope and purpose. Modern JavaScript (ES6+) also introduces features that make code more readable, maintainable, and efficient.

1. var

- var is function-scoped and can be re-declared and reassigned.
- It is generally avoided in modern JavaScript because of its unpredictable scoping behavior.

Example:

```
var cartItems = 0;
```

2. let

- let is block-scoped and allows reassignment.
- It is used for variables whose values may change during program execution.

Example:

```
let quantity = 2;
```

3. const

- const is block-scoped and cannot be reassigned after initialization.
- It is used for values that should remain constant throughout the program.

Example: `const gstRate = 0.18;`

Modern ES6 Features

1. Template Literals

- Template literals are enclosed within backticks ( ).
- They allow embedding variables and expressions using ${}.
- They also support multi-line strings.

Example:

```
console.log(`Welcome, ${customerName}!`);
```

2. Destructuring

- Destructuring is used to extract values from arrays or object properties into separate variables.
- It improves code readability and reduces repetitive code.

Example:

```
const product = {
 name: "Laptop",
```

price: 50000

};

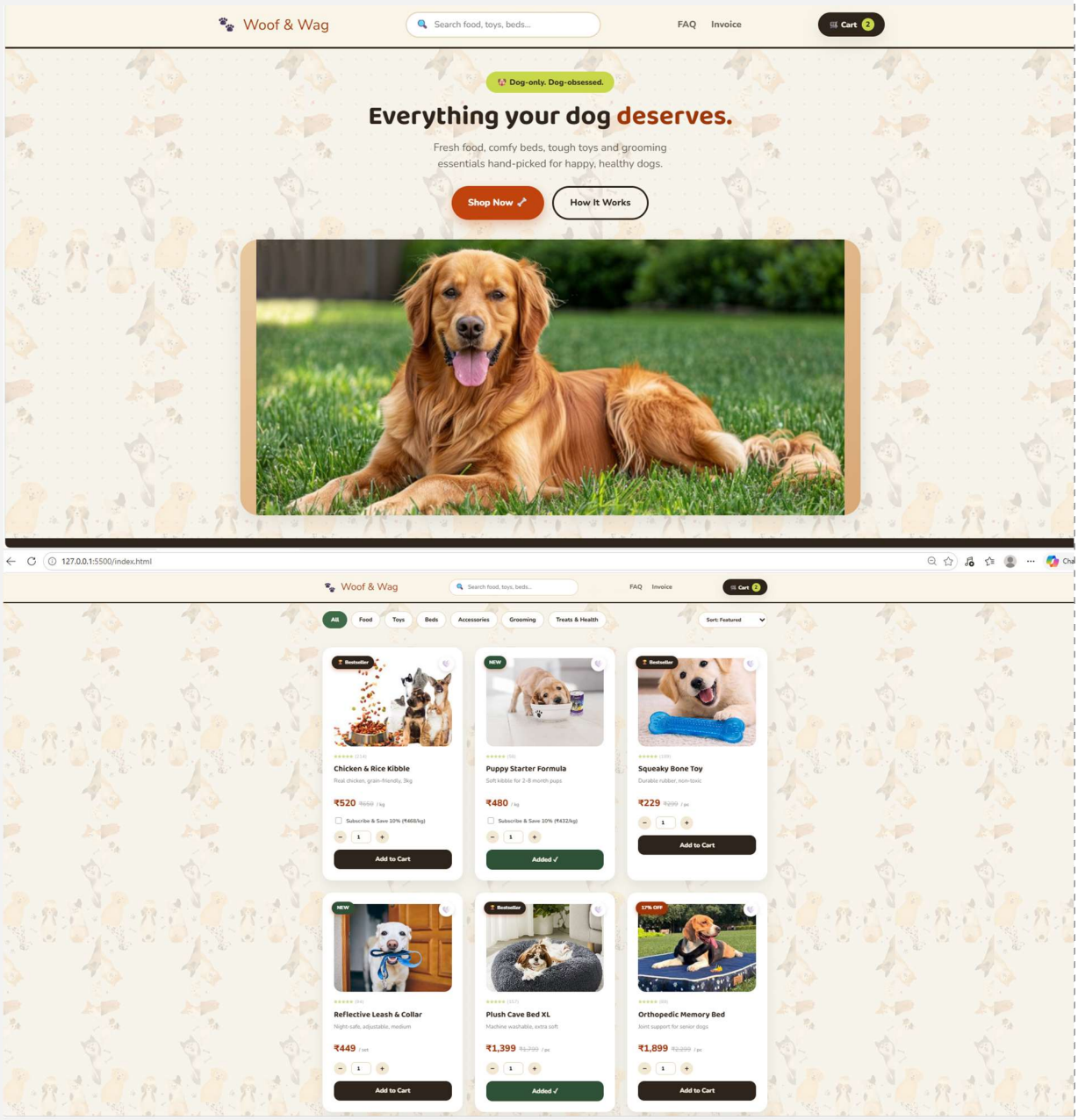
const { name, price } = product;

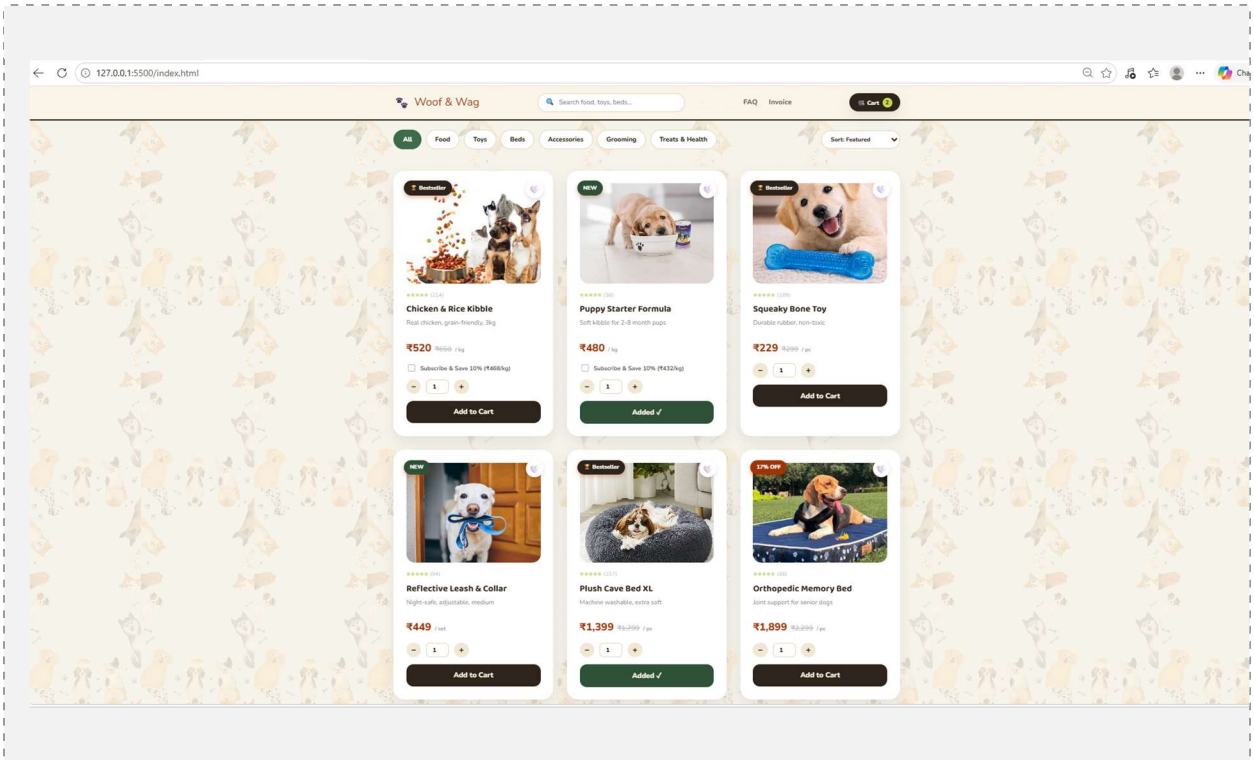
Aim of the CASE STUDY

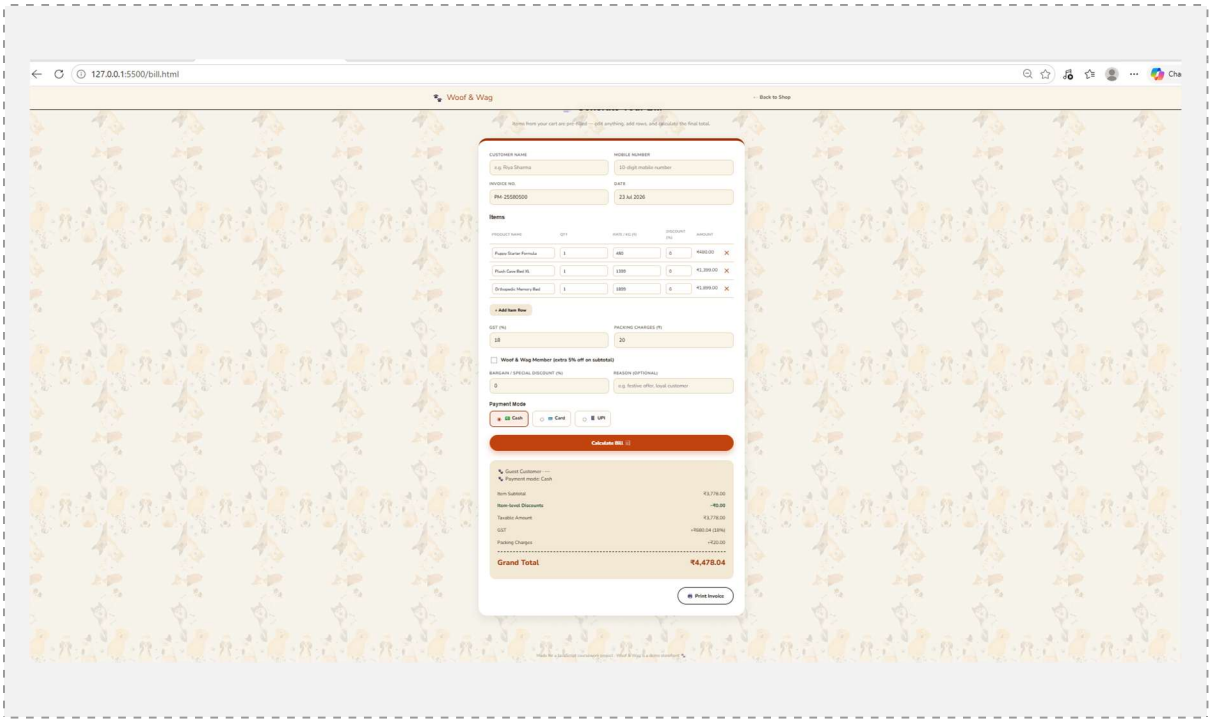
This experiment demonstrates the use of var, let, const, template literals, and object destructuring by developing a case study of simple Online Shopping Website. The webpage accepts customer details and product information, processes the purchase details, calculates the total amount and GST, and dynamically displays an order summary. Template literals are used to generate the order details, while destructuring is used to extract product information efficiently.

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File path - "C:\Users\ASUS\OneDrive\Desktop\symbi\Sem 5\JavaScript\Practical 2\Case study 2\Dog products website"


Woof & Wag
Search food, toys, beds...
FAQ
Invoice
Cart 2
Dog-only. Dog-obsessed.
Everything your dog deserves.
Fresh food, comfy beds, tough toys and grooming essentials hand-picked for happy, healthy dogs.
Shop Now
How It Works
127.0.0.1:5500/index.html
Woof & Wag
Search food, toys, beds...
FAQ
Invoice
Cart 2
All
Food
Toys
Beds
Accessories
Grooming
Treats & Health
Sort: Featured
Bestseller
Chicken & Rice Kibble
Real chicken, grain-friendly, 3kg
₹520 ₹650 / kg
Subscribe & Save 10% (₹468/kg)
Add to Cart
NEW
Puppy Starter Formula
Soft kibble for 2–8 month pups
₹480 / kg
Subscribe & Save 10% (₹432/kg)
Added ✓
Bestseller
Squeaky Bone Toy
Durable rubber, non-toxic
₹229 ₹299 / pc
Add to Cart
NEW
Reflective Leash & Collar
Night-safe, adjustable, medium
₹449 / set
Add to Cart
Bestseller
Plush Cave Bed XL
Machine washable, extra soft
₹1,399 ₹1,799 / pc
Added ✓
17% OFF
Orthopedic Memory Bed
Joint support for senior dogs
₹1,899 ₹2,299 / pc
Add to Cart





Output:

- A webpage titled "Online Shopping Website for PENS" is displayed.
- The user enters the Customer Name, Product Name, Quantity, and Price per Unit.
- After clicking the submit button, an Order Summary section is displayed dynamically.

- The webpage shows the Customer Name, Product Name, Quantity, Price per Unit, Total Amount, GST (18%), and Final Amount.
- The order summary is generated using template literals, and object destructuring is used to retrieve product details and then user can print the bill.

Result/Conclusion:

The experiment was successfully completed. The concepts of var, let, and const were implemented to understand variable declaration and scope. Modern JavaScript features such as template literals and destructuring were successfully used while developing an Online Shopping Website. The application accepted user input, processed product and customer details, calculated the total amount and GST, and displayed a dynamically generated order summary, demonstrating the practical implementation of ES6 features in web development and prints a bill at last.